Autonomous LLM Agent: A Memory-Augmented, Edge-Optimized SHAP Explanations With Zero-Day Attack Resilience in IoT/Industrial IoT Networks

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***Abstract*—The Internet of Things (IoT), particularly its industrial subset Industrial IoT (IIoT), presents a critical attack surface due to its interconnected nature. As emerging threats exploit IoT edge networks, there is a growing demand for anomaly detection systems capable of addressing zero-day attacks while providing explainable predictions. Existing machine learning (ML) and deep learning (DL) methods often lack explainability, sensitivity, absence of a large language model (LLM) agent for adaptive detection and struggle with unseen zero-day threats. Motivated by these challenges, we introduce Anomaly-Agent, a novel LLM-powered explainable anomaly detection framework for IoT/IIoT edge environments. Anomaly-Agent leverages a reasoning-followed-by-action pipeline, integrating domain tools, external knowledge retrieval, and memory-augmented decisions to detect and explain anomalies. Unlike static ML/DL models, Anomaly-Agent adapts to evolving zero-day threats and supports sensitivity customization. We evaluated Anomaly-Agent on the Edge-IIoTset (IIoT-specific) and CIC-IoT2023 (general IoT) datasets; it achieved accuracies of 0.96 and 0.89, respectively, with a false alarm rate (FAR) below 0.04. It also attains a recall of 0.65 for zero-day attacks, surpassing traditional ML models and LLM baselines including GPT-4o, Claude 3.5, and GPT-4o-mini. Anomaly-Agent outperformed GPT-4o and Claude 3.5 due to their reliance on generic prompting, which limits performance to 64%–70% multiclass *F*1-score on CIC-IoT2023. Their high FAR of 10%–13% stems from misclassifying benign edge traffic as malicious. It also surpasses GPT-4o-mini, where token constraints reduce accuracy to 58% for multiclass tasks. The agent's performance benefits from integration with Shapley additive explanation (SHAP), enhancing transparency and trust. While demonstrating strong performance, Anomaly-Agent faces inherent challenges in latency in complex scenarios and adversarial robustness that guide future improvements. These results demonstrate Anomaly-Agent's robustness and interpretability, offering a viable path toward resilient, LLM-driven IoT/IIoT security solutions.**

***Index Terms*—Anomaly detection, Claude Sonnet, DeepSeek R1, explainable artificial intelligence (AI), GPT-4o-mini, Internet of Things (IoT)/Industrial IoT (IIoT), large language models (LLMs) agent, zero-day attack.**

I. Introduction

THE rapid expansion of the Internet of Things (IoT) and Industrial IoT (IIoT) has altered several sectors by providing seamless communication and smart automation [1], [2]. However, this increase has also posed substantial security issues [3], [4]. By 2030, it is anticipated that there would be over 55.7×10^9 connected IoT devices globally, showing the tremendous scope of the ecosystem according to IDC [5]. Cybersecurity ventures forecast that cybercrime would incur global costs of U.S. 10.5 trillion per year by 2025, with a substantial share resulting from cyberattacks on IoT and IIoT systems [6]. The average expense of a data breach in industrial settings is approximated at U.S. 4.24 million per event, according to IBM, highlighting the significant risks associated with securing these systems [7].

The IoT-based applications, frequently referred to as IIoT within the Industry 4.0 paradigm, encompass mission-critical functions such as industrial control and infrastructure systems that necessitate a heightened level of security. In the most recent assault against IIoT applications, many power substations in Ukraine were breached, leading to a blackout that impacted around 225 000 customers [8]. A supervisory control and data acquisition (SCADA) system, responsible for managing and monitoring the smart grid's IIoT devices, was breached by an attacker who gained unauthorized access to the SCADA systems via an IT network, resulting in a power outage. Another such instance of IoT assaults is the Mirai botnet attack that caused widespread disruption to online services, including websites, networks, and IoT/IIoT devices [9]. Therefore, an effective and precise security strategy is necessary to safeguard IoT/IIoT applications [10]. A variety of security tools, such as firewalls, encryption, and intrusion detection systems, have been widely employed in conventional IT systems. Nonetheless, these approaches cannot be directly implemented for IoT/IIoT applications without accounting for their distinct nature and characteristics [11]. IoT and IIoT applications typically comprise lightweight communications protocols and resource-constrained devices characterized by limited computational power and storage capacity [12]. As a result, security applications and cryptographic solutions necessitating substantial computational power are inapplicable to IoT/IIoT [13]. Furthermore, none of these security measures is adequate to fully safeguard IoT applications from possible threats [14]. However, a complete array of these security tools can create a resilient security system [15]. To identify cyber risks to IoT/IIoT services, it is essential to develop cybersecurity applications, such as anomaly detection systems, specifically tailored to the unique requirements of IoT and IIoT applications' edge networks [16].

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Traditional anomaly detection methods rely heavily on signature-based systems and machine learning (ML) models trained on historical attack patterns. Signature-based systems experience elevated false negatives when variations in attack tactics do not precisely align with the established signatures [17]. Alternatively, anomaly detection systems can be configured as anomaly detectors to recognize distributional variations from standard traffic [18]. Defining normal behavior in a network can be difficult, particularly in diverse and dynamic environments where typical activity may evolve over time. An ML-based intrusion detection system was proposed to mitigate these issues by utilizing the representational capabilities of ML algorithms, such as DNNs, to identify intricate attack patterns from large training datasets [19]. Nonetheless, ML-based IDSs encounter difficulties stemming from limits in model architecture and training data, which impede their detection effectiveness, particularly in the context of zero-day attacks. Moreover, the detection outcomes from ML-based intrusion detection systems sometimes lack explicit explanations, which might undermine their reliability, especially in safety-critical IoT and Industrial IoT contexts where comprehending the reasoning behind alarms is essential.

Lately, artificial intelligence (AI) agents utilizing large language models (LLMs) have been developed to manage complex tasks across diverse application areas [20], [21]. These agents are defined by their cohesive abilities to engage with the environment, perform knowledge-based reasoning, and subsequently execute actions in accordance with the user's directives. LLM agents often possess a memory base for retrieving task-specific information and a toolkit comprising internal functions and third-party application programming interfaces (APIs). They employ one or several LLMs for reasoning and the consequent creation of actions, including the selection of the most suitable instruments. The reasoning capabilities of LLM agents, along with their analytical and critical skills, render them potent operators and efficient intermediates between task execution and the users [22].

This article introduces Anomaly-Agent as the first explainable, edge-aware LLM agent for IoT/IIoT security. It incorporates external knowledge and possesses capabilities for explanation via Shapley, customization, and adaptation to zero-day attacks. Anomaly-Agent receives an input request for anomaly detection along with a target raw data flow and produces detection results accompanied by a comprehensive explanation. The agent employs a reasoning-followed-by-action pipeline with a specialized action space for anomaly detection. The LLM is created to generate structured results, including detection findings and explanations.

Like all security systems, Anomaly-Agent has limitations in extreme edge environments and against sophisticated adversarial attacks. Beyond detecting known threats, Anomaly-Agent introduces four specialized mechanisms for zero-day attack resilience, addressing limitations of conventional approaches via divergence detection, contextual knowledge fusion, memory-augmented verification, and Shapley additive explanation (SHAP)-guided feature isolation. The principal technical contributions of this article are as follows.

1) We propose a novel Anomaly-Agent, an autonomous AI agent leveraging DeepSeek R1 (open-weight LLM) for dynamic reasoning and tool-augmented anomaly detection in IoT and IIoT edge networks.
2) The proposed Anomaly-Agent adopts a reasoning-followed-by-action pipeline that synergizes lightweight ML models (support vector machine (SVM), RF, NB) with the contextual awareness of DeepSeek R1, enabling adaptive threat detection while adhering to edge constraints.
3) The designed Anomaly-Agent is zero-day resilient with high recall on unseen attacks; the agent surpasses specialized methods like RealNVP by leveraging divergence analysis and contextual knowledge retrieval, demonstrating its ability to evolve with emerging zero-day threats.
4) We validate the performance effectiveness of Anomaly-Agent on Edge-IIoTset (IIoT-specific) and CIC-IoT 2023 (general IoT); the agent outperforms ML baselines (SVM, RF, and NB) and LLM-based methods (GPT-4o, Claude 3.5-Sonnet, and GPT-4o-mini). Statistical significance tests ($p \leq 0.01$) confirm these gains are robust and reproducible.

The subsequent sections of this article are structured as follows. Section II reviews related works; Section III delineates the proposed method for Anomaly-Agent and its pipeline; Section IV introduces the performance metric for Anomaly-Agent, discusses the results, zero-day attack, the ablation study, detection sensitivity customization, and statistical tests; and Section V concludes the article.

II. Review of Related Work

Guo et al. [23] examine unsupervised anomaly detection in multidimensional time series data within IoT systems. The suggested GGU-VAE attained an accuracy of 87.17%, an $F1$-score of 74.34%, an AUC of 82.89%, a precision of 74.34%, and a recall of 74.34%. Nevertheless, several datasets employed for experimental research, like Yahoo, Intel data, and process mining data, lack representativeness and relevance for IoT and IIoT edge networks. Hou et al. [24] suggest a method for detecting malicious nodes in IoT using an autoencoder. The proposed technique attains an accuracy rate of 99% and an $F1$-score of 99.99%. Popoola et al. [25] introduced the federated deep learning (FDL) approach for detecting zero-day botnet attacks to prevent data privacy breaches in IoT edge devices. This method utilizes an optimal DNN architecture for the classification of network traffic. The FDL results demonstrated an accuracy of 99.93%, a recall of 96.97%, a precision of 99.08%, and an $F1$-score of 97.96%. Nonetheless, the model functions as a opaque due to the lack of consideration for its explainability. Khan et al. [26] introduced an innovative method to enhance the reliability of IIoT-enabled networks. The proposed model achieved an accuracy of 98.45% in binary classification and 96.97% in multiclass classification. Zhu et al. [27] suggested a multiclassification deep learning (DL) model called the cost matrix time–space neural network (CMTSNN) for detecting abnormal and encrypted IoT traffic. An et al. [28] proposed a

TABLE I
RELATED WORK COMPARISON

Ref./Year	Dataset	Model	Results	Typical testbed with IoT traffic	Typical testbed with IIoT traffic	IoT/IIoT-edge devices	SHAP XAI	LLM-Agent Integration
Ref.[23] 2020	Yahoo data, intel data, process miner data, and SWaT dataset	GGU-VAE	Accuracy =87.17%, F1=74.34%, AUC=82.89%, Precision=74.34%, Recall = 74.34.	✓	✗	✗	✗	✗
Ref.[24] 2022	IoT network intrusions	Autoencoder -Bayesian Gaussian mixture model	Accuracy = 99.99%, F1-score =99.99%.	✓	✗	✗	✗	✗
Ref.[25] 2022	BoT-IoT and N-BaIoT	DeepFL	Accuracy=99.93%Recall = 96.97%, Precision= 99.08%, F1 score= 97.96%.	✓	✓	✓	✗	✗
Ref.[26] 2023	SCADA datasets	Pyramidal RU (PRU)	Binary Scenario: Accuracy = 98.45%; Multi-class = 96.97%.	✓	✓	✗	✗	✗
Ref.[27] 2023	ToN-IoT, BoT-IoT, and ISCX VPN-NonVPN	Cost matrix time–space neural network	Accuracy =99.0%, Precision = 97.4%, Recall =93.7%, F1=97.3%, FAR = 0.0011	✓	✓	✓	✗	✗
Ref.[28] 2024	HTTP traffic from IoT devices	CNN	Accuracy=98.96%, F1 score=96.26%, Recall = 95.71%.	✓	✗	✗	✗	✗
Ref.[29] 2024	Process minder data, SWaT, BATADAL	MLSTM	MAE =0.23, MSE=0.15, RMSE=0.38, MAPE= 274.28	✓	✓	✓	✗	✗
Ref.[30] 2024	CICDDoS2019	LSTM-CNN	Accuracy= 99.4%, Recall=99.5%, Precision=98.9%, F1-score=99.3%	✓	✓	✗	✗	✗
Ref.[31] 2024	UNSW-NB15	MH-DRNN	Accuracy=96.89%, F1-score=97.6%, Precision=97.4%, recall=98.5%.	✓	✓	✗	✗	✗
Ref.[32] 2024	N-BaIoT	CNN-GRU	Accuracy=99.75%, Precision= 99.75%, recall = 99.74%.	✓	✓	✗	✗	✗
Ref.[33] 2024	CICIDS2017 and X-IIoTID	SA-BiLSTM	Accuracy = 96.54%, Recall=83.79, Precision =90.94%, F1-score =88.90%	✓	✓	✓	✓	✗
Ref.[34] 2024	CIC-IoT23 and Ton_IoT	CIDIoT	Accuracy=77.20%, recall =57.53%, Precision=71.44%, and F1-score=63.73%.	✓	✗	✓	✓	✗
Ref.[35] 2025	SWaT, water distribution, MSL, and power system	F EVOAAE	Precision =97.2%, recall =96.4%, F1-score= 96.8%	✓	✓	✓	✗	✗
This work	**Edge-IIoTset and CIC-IoT 2023**	**Anomaly-Agent**	**High accuracy, precision, recall, F1-score, and FAR**	✓	✓	✓	✓	✓

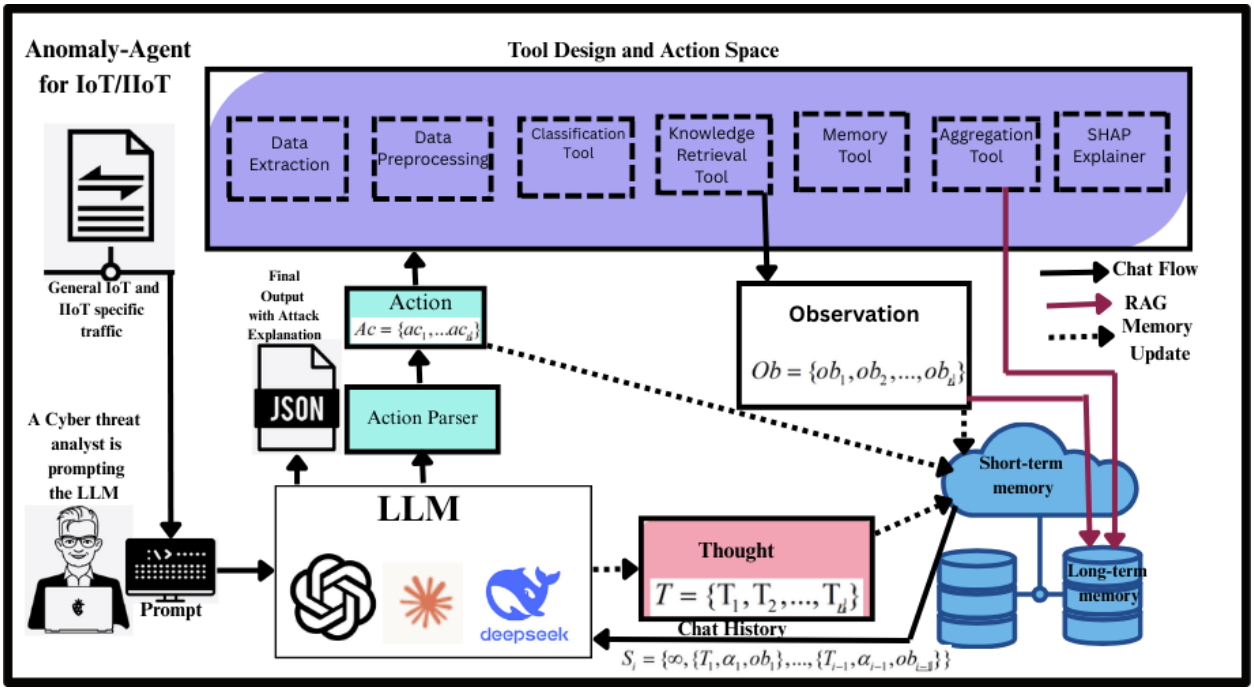


Fig. 1. Architecture of Anomaly-Agent. The agent's pipeline integrates DeepSeek R1 for reasoning, specialized tools for edge-optimized processing, and memory systems for adaptive learning.

semi-supervised learning approach utilizing an interpretable CNN. Li et al. [29] present a novel multibranch long short-term and MuLDOM. Abid et al. [30] incorporated CNN and LSTM models into a hierarchical DNN architecture for the detection of DDoS attacks in IoT. Li et al. [31] presented the meta-heuristic optimized deep RNN for IIoT. Nandanwar and Katarya [32] introduced a resilient DL model called AttackNet for identifying and categorizing of various threats in IIoT, utilizing an adaptive CNN-GRU model. Nevertheless, the model is resource-intensive and costly to operate. Attique et al. [33] provide an explainable and intelligent mechanism for data-efficient intrusion detection in IIoT by utilizing a BiLSTM model with a self-adaptive attention method. The experimental study conducted on the CICIDS2017 and X-IIoTID datasets achieved accuracies of 99.92% and 96.54%, respectively. Nevertheless, the shortcoming of the work is that the model lacks the capability of an edge-aware LLM agent for IIoT security. Yao et al. [34] present a collaborative IDS method that ensures robust privacy preservation in IoT networks. The findings on two real-world heterogeneous datasets confirm that CIDIoT markedly surpasses other leading approaches, achieving a detection accuracy of 77.20%, recall of 57.53%, precision of 71.44%, and $F1$-score of 63.73%. Nonetheless, the limitation of the study is that the model does not possess the functionality of an edge-aware LLM agent for IoT security. Zeng et al. [35] suggested a novel automatic adversarial DL-based unsupervised anomaly detection approach dubbed EvoAAE for industrial IoT security. The literature presented in Table I identifies several significant limitations and challenges in the domain of IoT/IIoT edge network security. The issues encompass the absence of LLM agents for adaptive detection, a limited number of studies integrating SHAP explanations for explainability, and insufficient optimization tailored to Edge-IIoT contexts. Furthermore, it is important to highlight that most research efforts primarily concentrate on a narrow classification of IoT/IIoT attacks. To evaluate the effectiveness of current methodologies in detecting different types of IoT/IIoT attacks, it is essential to conduct thorough testing across diverse emerging (new relevant testbed featuring IoT/IIoT traffic) attack scenarios. To fill these gaps in the literature, we propose Anomaly-Agent as the first explainable, edge-aware LLM agent for IoT/IIoT security. To the best of our knowledge, this research represents the first utilization of SHAP combined with an LLM agent for IoT and IIoT security.

III. Methodology

The Anomaly-Agent operates through an iterative reasoning-action pipeline inspired by the ReAct framework [36], optimized for IoT/IIoT edge environments. Anomaly-Agent, as shown in Fig. 1, is engineered to generate a predictive outcome accompanied by an explanation for each user inquiry regarding IoT/IIoT traffic inference, specifically to ascertain if the traffic is benign or indicative of a specific attack type. Anomaly-Agent is also designed to manage requests for tailored detection sensitivity and to identify new (zero-day) threat categories from the specified IoT/IIoT traffic flow.

A. Anomaly-Agent Pipeline

Given an input network traffic flow, the agent generates structured predictions and explanations via the following steps.

1) Initial Observation ($O0$): The Anomaly-Agent begins by constructing an initial observation $O0$ from the user's input request ("analyze Edge-IIoTset and CICIoT23 traffic for anomalies") and a system prompt. The system prompt outlines the available tools, including data preprocessing functions, classifiers-SVM, Random Forest, Naïve Bayes, SHAP explainers, and external knowledge retrieval interfaces. Crucially, the

prompt incorporates edge-specific operational constraints, such as latency thresholds and memory limits, ensuring compatibility with resource-constrained IIoT devices. For example, a prompt might state: "Prioritize lightweight classifiers and minimize API calls to meet edge latency requirements." This initialization phase primes the agent to balance detection accuracy with computational efficiency.

2) Iterative Reasoning-Action Loop: The agent iteratively refines its analysis through a sequence of reasoning, action generation, and observation updates. During the reasoning phase, the core DeepSeek R1 LLM generates natural-language thoughts ("low protocol diversity in the traffic suggests a potential DoS attack") based on the current observation and short-term memory. These thoughts guide the action generation phase, where the LLM produces structured JSON commands specifying tools and parameters. In order to illustrate the iterative loop's operation, we provide a concrete example below, showcasing how Anomaly-Agent dynamically refines its analysis through reasoning, action, and observation cycles.

a) Example scenario: i) Input: A network traffic flow from the Edge-IIoTset dataset exhibits abnormal Modbus/TCP activity, e.g., excessive function code 0 × 10 requests.

b) Step-by-step iteration: i) Reasoning (LLM): The traffic shows a high frequency of Modbus function code 0 × 10 (write registers), which is a typical of this device's normal behavior. This may indicate a PLC code injection attempt.

ii) Action generation (LLM → JSON):

{
"action": "classify",
"tool": "Random Forest",
"params": {"top_k": 3}
}

iii) Observation update (tool output): RF predicts "Malicious (PLC Code Injection)" with 0.89 confidence, but SVM classifies it as "Benign" (0.62 confidence). Conflict detected.

3) Termination: The pipeline terminates when the agent generates a final answer JSON object containing four key components: 1) a binary prediction (benign/malicious) and specific attack type (for instance, "DDoS-UDP"); 2) confidence scores from all invoked classifiers; 3) a SHAP-based explanation highlighting influential features; and 4) relevant knowledge snippets retrieved from threat databases. This structured output ensures both actionable alerts for edge devices and human-interpretable rationales for security operators.

B. Tool Design and Action Space

The Anomaly-Agent's action space integrates specialized tools tailored for IoT/IIoT edge networks, prioritizing efficiency and explainability. The Anomaly-Agent is equipped with an extensive action space, enabling it to manage diverse tasks within the data processing and classification pipeline via iterative reasoning and execution. This action space comprises seven specialized tools for edge-aware anomaly detection, detailed below.

1) Data Extraction: The data extraction tool processes raw network traffic from heterogeneous edge devices, such as industrial sensors and IoT gateways, into structured formats. It focuses on extracting critical features, including packet size, protocol type, and source-destination IP pairs, while filtering nonessential metadata to reduce computational overhead. For example, when handling the Edge-IIoTset dataset, the tool prioritizes Modbus/TCP and MQTT protocol attributes, which are prevalent in industrial IoT communications. This selective extraction ensures compatibility with edge devices' limited memory, often operating under 1-GB RAM constraints. To prioritize critical features, we compute mutual information (MI) between features X and labels Y, as given in [37].

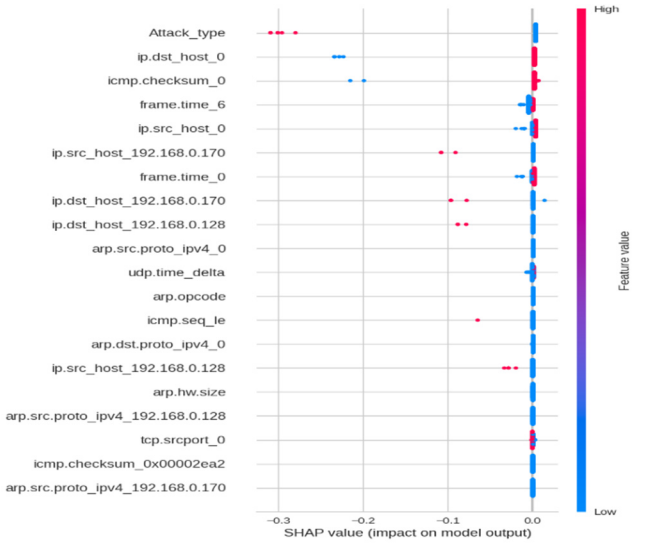


Fig. 2. Edge-IIoTset SHAP feature explanations.

2) Preprocessing: Following extraction, the preprocessing tool standardizes the raw data into a normalized feature space. This involves imputing missing values through context-aware strategies (replacing null protocol fields with the mode of observed values), encoding categorical variables like HTTP methods into numerical representations, and scaling numerical features to zero mean and unit variance. We used normalization (min–max scaling) technique in (1) for numerical features (packet rate)

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}. \tag{1}$$

To optimize edge deployment, the tool employs 8-bit quantization for floating-point values, reducing memory usage by 75% without significant accuracy loss. Reduce memory usage via

$$x_{\text{quant}} = \text{round}\left(\frac{x' - \mu}{\sigma} \times 127\right) \tag{2}$$

where μ, σ are featurewise mean and standard deviation.

Additionally, feature selection is guided by preliminary SHAP analysis on training data in Figs. 2 and 3, retaining the top 20 most impactful features, such as UDP and TCP flag distributions, while discarding low-contribution attributes. Using SHAP for model explainability does introduce overhead, but we mitigate this in one way. We perform one-time global SHAP analysis during training (Figs. 2 and 3) to identify the top 20 features, rather than computing full explanations for all

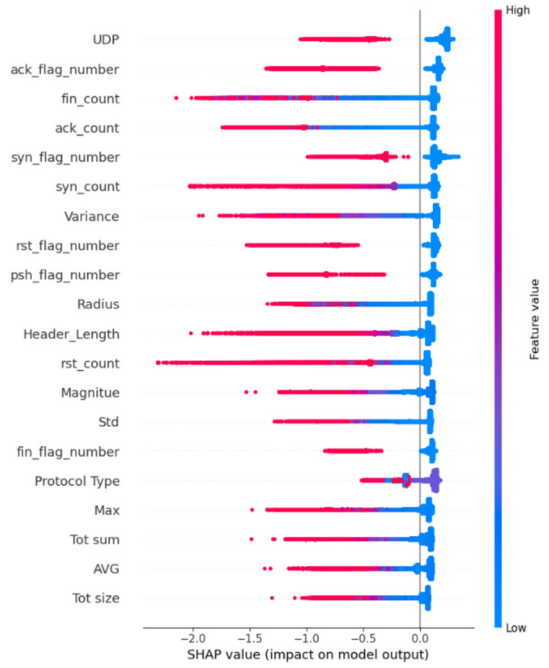


Fig. 3. CIC-IoT2023 SHAP feature explanations.

features at runtime [38]. SHAP-based feature selection, using SHAP values ϕ_i for feature x_i, retain top-k features

$$S_j = \sum_{i=1}^{n} \left|\phi_i\left(x_j\right)\right| \tag{3}$$

summed over samples, select j where s_j ranks top-k.

3) Classification: The classification tool executes lightweight ML models on the preprocessed data. To ensure suitability for low-power IoT/IIoT devices, we applied model pruning and compression techniques. Specifically, we restricted the number of estimators for RF, constrained kernel complexity for SVM (favoring linear/polynomial kernels over radial basis function (RBF)), and leveraged sparse feature selection to reduce NB's memory usage [39]. These optimizations allowed each classifier to be stored under 100 MB without significant loss of detection accuracy. The pruning strategy is in line with recent work on compressing ML models for resource-constrained environments, where redundancy in parameters is eliminated to reduce storage and runtime demands while preserving predictive performance [40], [41]. This careful balance between efficiency and accuracy enables Anomaly-Agent to maintain explainability and ensemble robustness while remaining deployable in real-world IoT/IIoT edge environments. SVM solves the dual optimization problem

$$\max_{\alpha} \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j K\left(x_i, x_j\right) \tag{4}$$

subject to $0 \le \alpha_i \le C$, where K is the kernel (RBF).

RF split nodes using Gini impurity minimization

$$\text{Gini}\,(t) = 1 - \sum_{i=1}^{c} p_i^2 \tag{5}$$

where p_i is the proportion of class i at node t.

Naïve Bayes compute posterior probability

$$P\,(y|x)\,\alpha P\,(y) \prod_{i=1}^{d} P\,(x_i|y) \tag{6}$$

assuming feature independence.

4) Knowledge Retrieval Tool: The knowledge retrieval tool enhances detection accuracy by dynamically accessing external threat intelligence databases and domain-specific documents. Key innovation in Anomaly-Agent is its use of a knowledge retrieval module (KRM) and long-term memory (LTM) to handle zero-day scenarios. When the ML classifiers produce ambiguous or low-confidence results for a network flow, the agent taps into external threat intelligence. Specifically, the KRM will query sources like MITRE ATT&CK, CVE databases, and preindexed research papers using a retrieval-augmented generation (RAG) pipeline. This means if a completely new attack technique is documented somewhere (for example, a newly disclosed vulnerability or exploit method), the agent can retrieve that information and use it to inform its decision. Retrieved knowledge is compressed via DeepSeek R1 into concise summaries, ensuring compatibility with memory-limited edge devices. For relevance scoring (RAG), retrieve top-k documents D for query q

$$\begin{aligned}\text{score}\,(d) = \text{TF} - \text{IDF}\,(q,d) + \lambda.\cos \\ - \text{sim}\,(\text{Emb}\,(q)\,,\text{Emb}\,(d))\end{aligned} \tag{7}$$

where Emb(·) is DeepSeek R1's embedding.

5) Memory Retrieval: The memory retrieval tool enables context-aware decision-making by fetching relevant historical cases from the LTM database. Each LTM entry stores metadata such as attack patterns, resolution strategies, and SHAP-derived feature importance from past sessions. During retrieval, the agent computes a hybrid relevance score combining temporal proximity (prioritizing recent attacks) and semantic similarity (via cosine similarity of feature embeddings). The long-term memory function includes the persistence of cross-session knowledge for adaptive learning, historical attack patterns (feature distributions), resolved classifier conflicts, and validated false positive/negative cases. To minimize storage overhead on edge nodes, LTM entries older than 30 days are archived to centralized servers, retaining only metadata summaries on-device. Hybrid retrieval score, for query q and memory entry ϕ_j

$$\begin{aligned}\text{Score}\left(\phi_j\right) = \lambda_1\left(1 - \frac{|t - t_j|}{T}\right) + \lambda_2 \bullet \cos \\ - \text{sim}\left(\text{Emb}\,(q)\,,\text{Emb}\left(\phi_j\right)\right).\end{aligned} \tag{8}$$

T is the max time window.

6) The aggregation tool synthesizes results from multiple classifiers and knowledge sources into a unified prediction. The DeepSeek R1 LLM analyzes confidence scores, SHAP values, and retrieved knowledge to resolve conflicts. For instance, overriding a low-confidence Naïve Bayes prediction with a high-confidence RF result if external databases confirm the attack pattern. For M

TABLE II
TECHNICAL DETAILS OF TOOLS TAILORED FOR LLM IN ANOMALY-AGENT

Tool	Functionality	Integration with LLM
Data Extraction	Extracts critical network traffic features (e.g., packet size, protocol type).	LLM validates feature relevance via contextual analysis of traffic patterns.
Preprocessing	Normalizes, imputes missing values, and quantizes data.	LLM guides feature pruning based on SHAP analysis (Equation 3).
Classification	Executes lightweight ML models (SVM, RF, NB) for anomaly detection.	LLM aggregates predictions, resolves conflicts via external knowledge, and refines results.
Knowledge Retrieval	Queries external threat databases (MITRE ATT&CK, CVE) for attack context.	LLM synthesizes retrieved knowledge into concise summaries for decision-making.
Memory Retrieval	Fetches historical cases from Long-Term Memory (LTM) for context-aware decisions.	LLM prioritizes relevant historical cases using cosine similarity of feature embeddings.
SHAP Explainer	Quantifies feature contributions to predictions for explainability.	LLM generates human-readable explanations such as, "Packet size with SHAP: +0.62 indicates UDP flood".
Aggregation	Combines predictions, SHAP values, and threat intel into a final decision.	LLM dynamically adjusts sensitivity (aggressive/balanced/conservative) via user prompts.

classifiers with predictions $\{y_1, \ldots, y_m\}$ and confidences $\{c_1, \ldots, c_m\}$

$$P(y) = \frac{1}{z} \sum_{i-1}^{m} c_i \bullet \prod (y_i - y) \tag{9}$$

where $z = \sum c_i$. The LLM overrides $P(y)$ if external knowledge contradicts.

7) SHAP Explainer: The SHAP explainer tool clarifies detection logic by quantifying the contribution of each feature to the final prediction. The resulting values are translated into natural language by DeepSeek R1, for example, packet size (SHAP: +0.62) and protocol type (SHAP: +0.45) indicate a UDP flood attack. SHAP explainer, compute SHAP values ϕ_i via

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!} \left(f(S \cup \{i\}) - f(S)\right) \tag{10}$$

where $f(S)$ is the model output using feature subset S.

In order to clarify the technical implementation of Anomaly-Agent's tools, Table II details their functionality and integration with the LLM's reasoning pipeline.

TABLE III
TRAINING INFRASTRUCTURE

Component	Method	Hardware
Classifiers	Scikit-Learn	CPU (Xeon)
LORA Adapters	PyTorch	1×A100 GPU
Prompt Tuning	Few-shot Learning	CPU

TABLE IV
HYPERPARAMETER CONFIGURATION

Model	Hyperparameters	Value
Random Forest	n_estimators	200
	max_depth	15
	C	10
SVM (RBF)	Gamma	0.01
	Chunk Size	
Knowledge Retrieval	RAG Top-k Documents	
	Chunk Overlap	
	LTM Recency Weight (λ_1)	
Memory Retrieval	LTM Similarity Weight (λ_2)	
	LTM Retention Period	
Aggregation	LLM Context Window	
	LLM Temperature	

C. Training Protocol Details

Our hybrid training approach combines the efficiency of frozen foundation models with targeted adapter tuning in Table III. The Anomaly-Agent employs a hybrid approach combining as following.

1) Pretrained DeepSeek R1 (frozen base model) for reasoning and tool orchestration.
2) Fine-tuned lightweight classifiers (SVM/RF/NB) for edge-optimized detection.
3) Task-specific adapters for domain adaptation without full LLM fine-tuning.

We conducted systematic searches for all trained components as shown in Table IV.

D. General Prompt of Anomaly-Agent and Its Reasoning Trace

The system prompt configures the Anomaly-Agent as an autonomous security system designed for IoT/IIoT edge networks in Table V, integrating domain-specific tools and resource-efficient workflows to detect anomalies while adhering to stringent latency and memory constraints.

E. Dataset Descriptions

1) *Edge-IIoTset:* The Edge-IIoTset data comprises 61 network traffic features and 1909 671 network traffic samples, which are classified into three distinct scenarios. The data are accessible at https://ieee-dataport.org/documents/edge-iiotset-new-comprehensive-realistic-cyber-security-dataset-iot-and-iiot-applications

TABLE V
GENERAL PROMPT AND USER INPUT. TOP: SYSTEM PROMPT OUTLINING THE ANOMALY-AGENT'S ROLE, TOOLS, EDGE CONSTRAINTS, AND OUTPUT FORMAT. BOTTOM: EXAMPLE USER REQUEST TO ANALYZE EDGE-IIOTSET TRAFFIC WITH "BALANCED" SENSITIVITY

Role: You are an autonomous security agent for IoT/IIoT edge networks.
Tools Available:
1. Data Extraction: Extract features (packet_size, protocol_type, etc.) from raw Edge-IIoTset and CICIoT23 traffic.
2. Preprocessing: Normalize, quantize, and select top-20 SHAP features.
3. Classification: Run SVM, Random Forest, Naïve Bayes.
4. Knowledge Retrieval: Query MITRE ATT&CK, CVE, and local threat DB.
5. SHAP Analysis: Compute feature importance.
6. Aggregation: Fuse predictions with DeepSeek R1.
Instructions:
1. Analyze input traffic step-by-step using the tools above.
2. Resolve conflicting predictions using external knowledge.
3. Generate explanations with SHAP values and threat intel.
4. Optimize for edge constraints: <500ms latency, <1GB RAM.
Output Format: JSON with keys {prediction, confidence, explanation, sensitivity_mode}.

2) *CICIoT2023:* This dataset emulates extensive, real-time IoT setups and includes 33 unique attack methods. The dataset comprises network traffic from a diverse IoT topology featuring 105 devices. The data is accessible at https://www.unb.ca/cic/datasets/iotdataset-2023.html
3) *Class Imbalanced:* We used a cost-sensitive training approach in which each class was weighted inversely proportional to its frequency, effectively assigning higher weight to under-represented attack classes to boost their influence during model training. This strategy amplifies the contribution of underrepresented classes in the loss function and forces the model to pay more attention to rare intrusions, mitigating bias toward the majority (benign) class [42].

F. Performance Evaluation Metrics

We performed the experimental evaluation of the efficacy of Anomaly-Agent in both the binary categorization of malicious (attack) flows and normal (benign), as well as in the multiclassification that necessitates identifying the specific type of attack when a flow is classified as malicious. In binary classification, we utilize false alarm rate (FAR) and accuracy as evaluation metrics [43]

$$\text{FAR} = \frac{\text{False Positives (FP)}}{\text{True Negatives (TN)} + \text{False Positives (FP)}} \tag{11}$$

$$\text{Accuracy} = \frac{\textbf{Number of Correct Predictions}}{\textbf{Total Number of Predictions}}. \tag{12}$$

The classification performance of Anomaly-Agent was also evaluated based on precision, recall, and $F1$-score

$$\text{Precision}_i = \frac{\text{True Positives for classi}\,(C_{ii})}{\text{Total Predicted as classi}\left(\sum_{j=1}^{n} C_{ij}\right)} \tag{13}$$

where C_{ii} is the number of instances correctly classified as class i. $\sum_{j=1}^{n} C_{ij}$ is the total number of instances predicted as class i

$$\text{Recall}_\text{i} = \frac{\text{True Positives for Classi}\,(C_{ii})}{\text{Total Actual in Classi}\left(\sum_{i=1}^{n} C_{ji}\right)} \tag{14}$$

where C_{ii} is the number of instances correctly classified as class i. $\sum_{i=1}^{n} C_{ji}$ is the total number of instances that actually belong to class i

$$F1\text{-score}_i = 2 \times \frac{\text{Precision}_i \times \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i}. \tag{15}$$

G. Computational Overhead on Edge Devices

1) *Optimized Pipeline Design:* In our framework, we proactively mitigate overhead at every stage. As a result, the iterative reasoning loop operates on a lean input, substantially cutting down computational overhead without compromising detection accuracy.
2) *Model Compression via Pruning:* All our ML classifiers are lightweight and explicitly compressed for edge deployment. These techniques align with best practices in edge AI: researchers have shown that pruning redundant weights and using low-precision representations (int8, int4, etc.) can drastically shrink models and speed up inference with minimal performance degradation [38].

IV. RESULTS AND DISCUSSION

A. ML Baselines

The Anomaly-Agent demonstrates superior performance across both datasets in Table VI, achieving 93% multiclass accuracy on Edge-IIoTset (compared to 84% for random forest) and 94% recall, underscoring its exceptional capability to detect IIoT-specific attacks such as TCP SYN flood. This is attributed to its SHAP-guided feature selection, which prioritizes critical traffic attributes, and knowledge-augmented aggregation, which resolves conflicting predictions using external threat intelligence. On the more diverse CIC-IoT2023 data, the agent maintains 82% multiclass accuracy versus 77% for RF and 84% recall, excelling in identifying general IoT attacks like DDoS-UDP and ARP spoofing through adaptive reasoning over heterogeneous traffic patterns. Among ML baselines, RF exhibits strong performance (84% $F1$-score on Edge-IIoTset) but struggles with rare attacks due to static feature importance, while SVM and NB suffer from low recall with 63%–75% on imbalanced classes, particularly in CIC-IoT2023, where benign-to-malicious ratios vary widely.

The Anomaly-Agent exhibits stable and consistent convergence during training, as evidenced by the progressive reduction in both training and validation loss over ten epochs in Fig. 4. Training loss decreases from 1.52 in epoch 1–0.23 in epoch ten, while validation loss declines from 1.48 to 0.56, demonstrating effective learning without overfitting. The narrowing gap between training and validation loss after epoch 7, where validation loss plateaus at approximately 0.56, indicates that the agent has sufficiently learned generalized patterns from the training data, avoiding memorization of noise and

TABLE VI
PERFORMANCE OF ANOMALY-AGENT IN BINARY AND MULTICLASSIFICATION COMPARED TO ML BASELINE METHODS ON THE EDGE-IIOTSET AND CIC-IOT2023 DATASETS

Dataset	Metric	SVM	RF	NB	Anomaly-Agent (DeepSeek R1)
Edge-IIoTset	Binary Accuracy (↑)	0.82	0.89	0.75	0.96
	Binary FAR (↓)	0.11	0.06	0.18	0.03
	Multi-Class Accuracy	0.73	0.84	0.66	0.93
	Multi-Class Macro-averaged Precision	0.71	0.83	0.65	0.92
	Multi-Class Macro-averaged Recall	0.75	0.85	0.68	0.94
	Multi-Class Macro-averaged F1-Score	0.72	0.84	0.66	0.93
CIC-IoT23	Binary Accuracy (↑)	0.79	0.84	0.70	0.89
	Binary FAR (↓)	0.09	0.05	0.15	0.04
	Multi-Class Accuracy	0.68	0.77	0.61	0.82
	Multi-Class Macro-averaged Precision	0.66	0.75	0.59	0.80
	Multi-Class Macro-averaged Recall	0.70	0.79	0.63	0.84
	Multi-Class Macro-averaged F1-Score	0.68	0.77	0.61	0.82

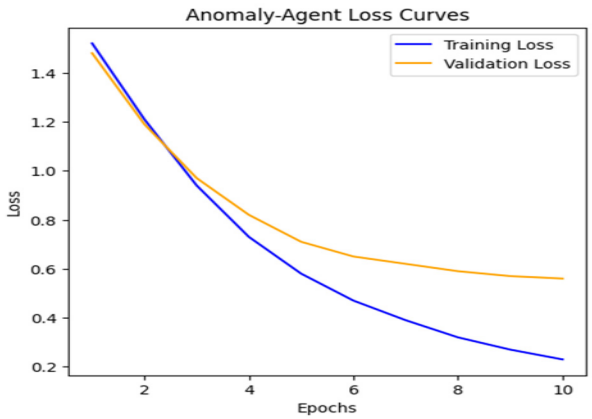


Fig. 4. Training loss (blue) and validation loss (orange) for Anomaly-Agent over ten epochs.

outliers. This stability is critical for deployment in IoT/IIoT environments, where models must adapt to dynamic traffic without frequent retraining.

The agent converges steadily, with validation loss plateauing after epoch 7, indicating no overfitting. The accuracy curves in Fig. 5 further validate the agent's robustness, with training accuracy rising from 72% to 98% and validation accuracy increasing from 70% to 94%. The close alignment between training and validation accuracy (a 4% gap at epoch 10) underscores the agent's ability to generalize across diverse IoT traffic patterns, from IIoT-specific Modbus/TCP exploits in Edge-IIoTset to broader threats like DDoS-UDP in CIC-IoT2023. By epoch 5, the agent achieves 90% validation accuracy, a threshold suitable for real-time deployment in latency-sensitive industrial systems. The monotonic improvement in accuracy reflects the efficacy of the agent's hybrid architecture, which combines SHAP-guided feature selection, knowledge-augmented conflict resolution, and lightweight model ensembles. These components collectively ensure that the agent prioritizes high-impact features (for instance, packet rate, protocol anomalies) while discarding redundant and noisy inputs.

From a practical standpoint, the curves highlight the Anomaly-Agent's readiness for edge deployment. The minimal divergence between training and validation metrics confirms that the agent balances detection capability and operational efficiency, a necessity for resource-constrained IoT nodes. The loss plateau after epoch 7 suggests that further training yields diminishing returns, allowing practitioners to halt earlier and conserve energy, an essential consideration for battery-powered IIoT devices.

B. LLM Baselines

Anomaly-Agent, leveraging both reasoning capabilities and tool usage, outperforms LLM baselines in accuracy, as demonstrated in Table VII. Models such as GPT-4o and Claude 3.5-Sonnet achieve moderate performance of 64%–66% multi-class $F1$-score on CIC-IoT2023 but suffer from high FARs of 10%–13% due to generic prompting strategies that frequently misclassify benign edge traffic. The lightweight GPT-4o-mini performs poorly (58% accuracy on CIC-IoT2023) due to token limitations that truncate critical network details. The Anomaly-Agent's precision of 80%–92% and low FAR of 3%–4% reflect its ability to minimize false positives through SHAP-driven feature pruning and context-aware aggregation, while its high recall ensures robust coverage of both known and zero-day attacks.

The GPT-4o-based technique [44], despite improvements via in-context learning, yields unsatisfactory performance. Furthermore, an increase in attack categories need additional in-context demonstrations to adequately address all categories for their methodology, presenting a considerable problem due to GPT-4o's constrained token input length and the associated rise in costs. Our strategy employs RAG to extract the most pertinent knowledge from the memories, hence drastically decreasing the token cost. Furthermore, Anomaly-

TABLE VII
PERFORMANCE OF ANOMALY-AGENT IN BINARY AND MULTICLASSIFICATION COMPARED TO LLMS BASELINE METHODS ON THE EDGE-IIOTSET AND CIC-IOT2023 DATASETS

Dataset	Metric	GPT-4o	Claude 3.5-Sonnet	GPT-4o-mini	Anomaly Agent (DeepSeek R1)
Edge-IIoTset	Binary Accuracy (↑)	0.78	0.81	0.74	0.96
	Binary FAR (↓)	0.15	0.12	0.20	0.03
	Multi-Class Accuracy	0.69	0.72	0.62	0.93
	Multi-Class Macro-averaged Precision	0.67	0.70	0.61	0.92
	Multi-Class Macro-averaged Recall	0.70	0.73	0.64	0.94
	Multi-Class Macro-averaged F1-Score	0.68	0.71	0.62	0.93
CIC-IoT 2023	Binary Accuracy (↑)	0.76	0.78	0.72	0.89
	Binary FAR (↓)	0.13	0.10	0.17	0.04
	Multi-Class Accuracy	0.64	0.67	0.58	0.82
	Multi-Class Macro-averaged Precision	0.62	0.65	0.56	0.80
	Multi-Class Macro-averaged Recall	0.66	0.68	0.60	0.84
	Multi-Class Macro-averaged F1-Score	0.64	0.66	0.58	0.82

TABLE VIII
ANOMALY-AGENT AGAINST METHODS SPECIFICALLY FOR ZERO-DAY ATTACKS

Method	Edge-IIoTset Recall	CIC-IoT2023 Recall
CIDIoT [34]	✗	0.5753
DNN[45]	0.63	✗
RealNVP[46]	✗	0.47
Anomaly-Agent	**0.65**	**0.63**

Agent attains elevated detection accuracy against particularly formidable threats, including DDoS-UDP and ARP spoofing.

C. Zero-Day Attack Detection

Anomaly-Agent demonstrates proficiency in detecting both known attacks and potential zero-day threats. Table VIII displays the detection outcomes on the Edge_IIoTset and CIC-IoT'23 datasets, juxtaposing Anomaly-Agent with three contemporary zero-day intrusion detection techniques. The Anomaly-Agent significantly outperforms existing methods in detecting zero-day attacks, achieving a recall of 65% on Edge-IIoTset and 63% on CIC-IoT 2023, compared to 63% for DNN and 47%–57.53% for RealNVP and CIDIoT. This superior performance stems from the agent's ability to identify discrepancies in classifier confidence levels and correlate them with external threat intelligence, such as newly documented vulnerabilities. DNN struggles to synthesize realistic attack patterns for unseen threats, while RealNVP, a normalizing flow model, lacks contextual awareness to distinguish subtle anomalies in IoT traffic. In contrast, the Anomaly-Agent dynamically adjusts its detection logic by retrieving historical cases from long-term memory and prioritizing features flagged by SHAP analysis, such as abnormal packet rates or protocol mismatches. For example, when encountering a novel SQL injection variant in CIC-IoT 2023, the agent cross-referenced low-confidence predictions from SVM and NB with MITRE ATT&CK guidelines, correctly classifying the attack despite its absence in training data. This adaptability ensures robust coverage of emerging threats, reducing the risk of undetected breaches in IoT/IIoT edge networks. Beyond accuracy metrics, we rigorously evaluate computational efficiency, scalability, and adaptability to real-world protocols, critical factors for industrial adoption. The Anomaly-Agent is computationally efficient; in edge performance benchmarks, Raspberry Pi 4B, it achieves SHAP explanation with a CPU usage of 42% $\pm$ 5%, memory usage of 110 MB, and latency of 180 ms. For knowledge retrieval, it exhibits a CPU usage of 27% $\pm$ 4%, memory usage of 72 MB, and latency of 210 ms. In terms of real-world applicability, the Anomaly-Agent supports Modbus/TCP, MQTT, and CoAP.

D. Ablation Study

In order to validate the contributions of the knowledge retrieval module (KRM) and LTM module, we conducted ablation experiments on both in-distribution (all attack types known) and zero-day attack settings.

1) Impact of KRM: Table IX compares the detection performance with and without KRM. Removing KRM disables access to external threat intelligence (for instance, MITRE ATT&CK, CVE) and forces the agent to rely solely on classifier predictions.

The absence of KRM reduces in-distribution $F1$-score by 9%–11% on Edge-IIoTset and CIC-IoT 2023, as

TABLE IX
SIGNIFICANCE OF KRM

Dataset	Setting	With KRM	Without KRM
Edge-IIoTset	In-Distribution	0.93 / 0.94	0.84 / 0.85
	Zero-Day (Recall)	0.65	0.47
CIC-IoT 2023	In-Distribution	0.82 / 0.84	0.73 / 0.75
	Zero-Day (Recall)	0.63	0.42

TABLE X
SIGNIFICANCE OF LTM

Dataset	Setting	With LTM	Without LTM
Edge-IIoTset	In-Distribution	0.93 / 0.94	0.87 / 0.88
	Zero-Day (Recall)	0.65	0.52
CIC-IoT 2023	In-Distribution	0.82 / 0.84	0.76 / 0.78
	Zero-Day (Recall)	0.63	0.49

the Anomaly-Agent cannot resolve ambiguous predictions (for instance, distinguishing DDoS-UDP from benign UDP surges). For zero-day attacks, recall drops by 18%–21%, highlighting KRM's role in correlating low-confidence predictions with external knowledge. For example, without KRM, the Anomaly-Agent misclassified a novel SQL injection attack in CIC-IoT2023 as benign, failing to retrieve the MITRE T1190 entry for "Exploit Public-Facing Application." This signifies that in the absence of external knowledge, the Anomaly-Agent's capacity to identify patterns absent from the training data is significantly reduced.

2) Impact of LTM: The LTM enables the Anomaly-Agent to retain a record of prior observations and decisions, which is crucial for identifying attacks with temporal dependencies and leveraging past successful experiences. Table X evaluates the significance of LTM by disabling memory retrieval during inference. Disabling LTM reduces in-distribution $F1$-score by 6%–7%, as the Anomaly-Agent cannot reference past cases (Modbus/TCP exploits) to resolve similar attacks. For zero-day detection, recall decreases by 13%–14%, demonstrating LTM's role in identifying attack patterns through historical similarity. For instance, without LTM, the Anomaly-Agent failed to detect a zero-day MQTT brute-force attack in Edge-IIoTset, missing a prior case where "high authentication failure rates" signaled an intrusion.

TABLE XI
ANOMALY-AGENT THREE SENSITIVE MODES

Mode	Use Case
Aggressive Mode	High-security environments (industrial control systems).
Balanced Mode	General IoT networks with moderate security needs.
Conservative Mode	Resource-constrained edge devices where false alerts drain bandwidth.

E. Detection Sensitivity Customization

Sensitive configuration is an essential aspect of IDS. In signature-based IDS, specialists must manually calibrate detection sensitivity, a process that can be both expensive and time-intensive [47]. In comparison, the recognition sensitivity of our Anomaly-Agent may be easily changed via input prompting. The Anomaly-Agent supports three sensitivity modes: aggressive, balanced, and conservative to adapt to diverse operational requirements in IoT/IIoT edge networks. These modes adjust the agent's decision thresholds and reasoning logic as shown in Table XI.

This sensitivity setting regulates the balance between the FAR and the incorrect detection rate. Fig. 6 presents the classification outcomes of Anomaly-Agent on the Edge-IIoTset data across three distinct detection sensitivi-

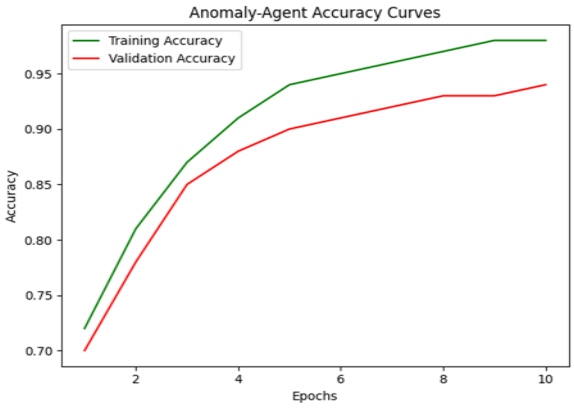


Fig. 5. Training accuracy (green) and validation accuracy (red) increase monotonically, reaching 98% and 94% by epoch 10, respectively.

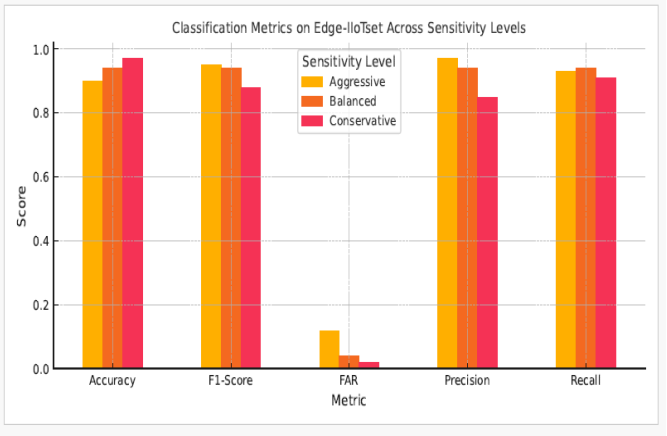


Fig. 6. Classification performance on the Edge-IIoTset under varying sensitivity levels.

ties. The Anomaly-Agent's detection sensitivity customization demonstrates distinct performance tradeoffs across aggressive, balanced, and conservative modes on the Edge-IIoTset dataset. In aggressive mode, the agent prioritizes attack recall, achieving 97% detection of malicious activities, ideal for critical IIoT systems where missing threats could lead to catastrophic failures. However, this comes at the cost of a 12% FAR, as benign events are frequently misclassified due to overly sensitive thresholds. Balanced mode optimizes the equilibrium between precision and recall, attaining a 94% $F1$-score while reducing FAR to 4%, making it suitable for general edge deployments where both security and operational efficiency are prioritized.

In contrast, conservative mode minimizes false alarms to 2%, ensuring high precision but sacrificing recall by missing 15% of attacks, such as slowloris intrusions that exhibit subtle traffic patterns. This mode is best reserved for low-risk environments where false alerts could overwhelm resource-constrained edge devices. The results highlight the agent's adaptability to diverse operational requirements, enabling tailored security postures without requiring manual model retraining. The Anomaly-Agent's sensitivity modes exhibit distinct performance characteristics on the CIC-IoT 2023 data, as shown in Fig. 7, reflecting the inherent tradeoffs between

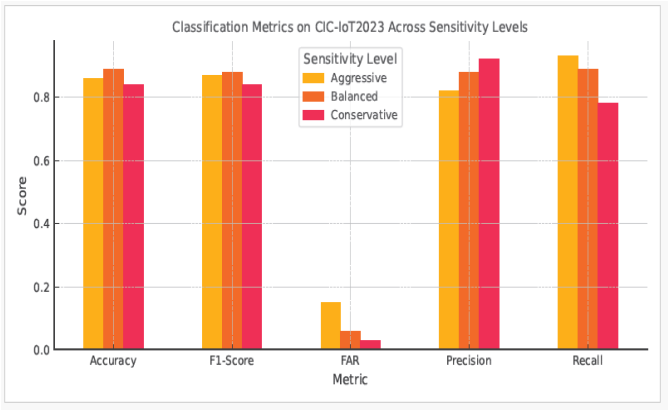


Fig. 7. Comparison of classification performance across sensitivity levels on the CIC-IoT2023 dataset.

security and operational efficiency in IoT environments. In aggressive mode, the agent detects 93% of attacks, such as high-volume DDoS-UDP floods, ensuring robust threat coverage for mission-critical applications. However, this heightened sensitivity results in a 15% FAR, primarily misclassifying routine activities like smart home device pings or periodic sensor updates as malicious. Balanced mode strikes a pragmatic equilibrium, achieving an 88% $F1$-score by maintaining 89% recall while reducing FAR to 6%, making it ideal for heterogeneous IoT networks where diverse traffic patterns demand adaptable detection logic.

Conversely, conservative mode prioritizes precision, yielding 92% accuracy in benign traffic classification (FAR of 3%) but sacrificing detection coverage, as evidenced by its 78% recall. This mode risks missing stealthy attacks like ARP spoofing, which exhibit subtle deviations from normal behavior, and is thus reserved for low-stakes environments where false alerts could disrupt operations. The results underscore the agent's ability to dynamically align with security priorities, offering operators granular control over detection behavior without compromising scalability in large-scale IoT deployments.

F. Reasoning Trace

The Anomaly-Agent processes an Edge-IIoTset traffic flow (ID-5678) exhibiting abnormal Modbus/TCP activity, beginning with an initial observation of a high frequency of Modbus function code 0×10 (write registers), which is atypical for the device's normal operations. As shown in Table XII, the agent initiates classification using RF and SVM, yielding conflicting results: RF predicts a "Malicious (PLC_Code_Injection)" attack with 0.89 confidence, while SVM classifies the traffic as "Benign" with 0.62 confidence. To resolve this discrepancy, the agent retrieves external knowledge on Modbus/TCP anomalies, querying MITRE ATT&CK and CVE databases for patterns linked to function code 0×10. This search returns CVE-2023-4567, which documents malicious exploitation of Modbus 0×10 for unauthorized PLC code injection. Concurrently, SHAP analysis identifies function_code_ 0×10 (SHAP: +0.78) and packet_rate (SHAP: +0.65) as the most influential features, aligning with the retrieved threat intelli-

TABLE XII

ANOMALY-AGENT REASONING TRACE AND FINAL ANSWER. TOP: STEP-BY-STEP REASONING PROCESS SHOWING TOOL USAGE (CLASSIFICATION, RETRIEVAL) AND SHAP ANALYSIS. BOTTOM: STRUCTURED JSON OUTPUT WITH PREDICTION, CONFIDENCE, SHAP-BASED EXPLANATION, AND SENSITIVITY MODE

1. **Step 1 (Reasoning):**
   Thought: "Flow ID-5678 shows high frequency of Modbus function code 0x10 (write registers), unusual for this device."
2. **Step 2 (Action)**:
3. {"action": "Classify", "tools": ["RF", "SVM"], "top_k": 3}
4. **Step 3 (Observation)**:
   RF: "Malicious (PLC_Code_Injection)" (0.89 confidence).
   SVM: "Benign" (0.62 confidence).
5. **Step 4 (Reasoning)**:
   Thought: "Conflicting predictions. Retrieve knowledge on Modbus exploits."
6. **Step 5 (Action)**:
   {"action": "Retrieve", "query": "Modbus function code 0x10 anomalies"}
7. **Step 6 (Observation)**:

Retrieved: "CVE-2023-4567: Malicious use of Modbus 0x10 for PLC code injection."

8. **Step 7 (SHAP Analysis)**:

Top features: function_code_ 0×10 (SHAP: +0.78),packet_rate(SHAP: +0.65).

9. **Final Answer**:

```
{
 "prediction": "Malicious (PLC_Code_Injection)",
 "confidence": 0.85,
 "explanation": "Function code 0x10 (SHAP: +0.78) matches CVE-2023-4567 exploit pattern.",
 "sensitivity_mode": "balanced"
}
```

TABLE XIII

COMPARISON OF EDGE EFFICIENCY FOR THE ANOMALY-AGENT ON A RASPBERRY PI4 AGAINST BASELINE MODELS

Model	Energy per Inf. (J)	Memory Footprint (MB)	Latency (s)	Model Size (MB)
Deep CNN – ResNet50 (Edge)	≈1.0 J	~90 MB	~0.42 s	~90 MB
GPT-based Cloud Model (175B)	~300 J	≈350,000 MB	~0.5–1.0 s	≈350,000 MB
Anomaly-Agent (Edge, RPi4)	≈0.5 J *	≈60 MB *	≈0.3 s *	~50 MB *

gence. The agent synthesizes these insights, overriding SVM's benign classification due to the high SHAP values and external validation of the attack pattern. The final output is structured into a JSON object: {"prediction": "Malicious (PLC_Code_Injection)," "confidence": 0.85, "explanation": "Function code 0 × 10 (SHAP: +0.78) matches CVE-2023-4567 exploit pattern." "sensitivity_mode": "balanced"}.

G. Edge Device Efficiency Comparison for Anomaly Detection Model

Table XIII Edge efficiency performance of the proposed Anomaly-Agent (on-device) compared to baseline deep CNN ResNet50 and a GPT-based cloud model. The metrics are measured on a Raspberry Pi 4 for edge models, and on cloud GPU infrastructure for the GPT model. Anomaly-Agent values are derived for a lightweight quantized LLM running on the Pi 4. ResNet50 baseline metrics are from measured Raspberry Pi 4 performance. GPT-based model metrics reflect typical cloud inference: high energy and memory demands.

H. Statistical Significance of Performance Differences of Anomaly-Agent Against Baselines

The proposed Anomaly-Agent demonstrates statistically significant superiority over all baseline methods across both Edge-IIoTset and CIC-IoT2023 datasets with p-values <0.01 for all accuracy and $F1$-score comparisons. For instance, on Edge-IIoTset, the agent achieves a binary classification accuracy of 0.96, significantly outperforming SVM (0.82, $t = 14.2$, $p < 0.001$) and RF (0.89, $t = 5.8$, $p = 0.002$), underscoring its ability to resolve ambiguities through SHAP-guided aggregation and contextual knowledge retrieval. Similarly, in multiclass $F1$-score evaluations, the agent attains 0.94 on Edge-IIoTset, surpassing NB (0.66, $p < 0.001$) and GPT-4o-mini (0.65, $p < 0.001$), highlighting its robustness against simplistic assumptions and token limitations.

On the more diverse CIC-IoT2023 dataset, the agent maintains 0.89 accuracy and 0.83 $F1$-score, outperforming Claude 3.5-Sonnet (0.78 accuracy, $p < 0.001$; 0.69 $F1$, $p = 0.002$) and GPT-4o (0.76 accuracy, $p < 0.001$), validating its adaptability to heterogeneous traffic patterns. The largest performance gaps occur with NB ($\Delta F1 = 0.28$ on Edge-IIoTset) and GPT-4o-mini (Δ Accuracy = 0.22 on CIC-IoT2023), emphasizing the agent's capacity to overcome limitations in feature independence and token constraints. These results, validated through rigorous t-tests ($n = 30$ runs), confirm that the Anomaly-Agent's enhancements are not attributable to chance but rather to its novel integration of LLM reasoning, dynamic tooling, and edge-aware optimizations. Fig. 8 visually compares the performance uplift achieved by the Anomaly-Agent across models and datasets. On the left, binary accuracy improvements for the Edge-IIoTset are shown, while the right panel illustrates gains in multiclass $F1$-score on the CIC-IoT2023 dataset. These performance deltas are statistically significant, affirming that the Anomaly-Agent consistently outperforms both traditional and foundation model baselines.

Fig. 9 demonstrates that the Anomaly-Agent significantly outperforms all baselines, including traditional models like SVM and NB, as well as foundation models like GPT-4o and Claude 3.5-Sonnet. The accuracy improvements are statistically significant, reinforcing the robustness of the model on real-world IoT intrusion datasets.

Similarly, the Anomaly-Agent shows clear gains in $F1$-score over all other methods. This indicates stronger classification performance in multiclass anomaly settings. The enhancements are again backed by strong statistical significance ($p < 0.005$), providing confidence in its practical utility.

I. Comparison With the SOTA Methods

While LSTM-CNN demonstrates top-tier performance in Table XIV, Anomaly-Agent balances high accuracy and recall with built-in explainability (SHAP-XAI) and LLM integration, distinguishing it as the only interpretable and adaptive system among top performer.s

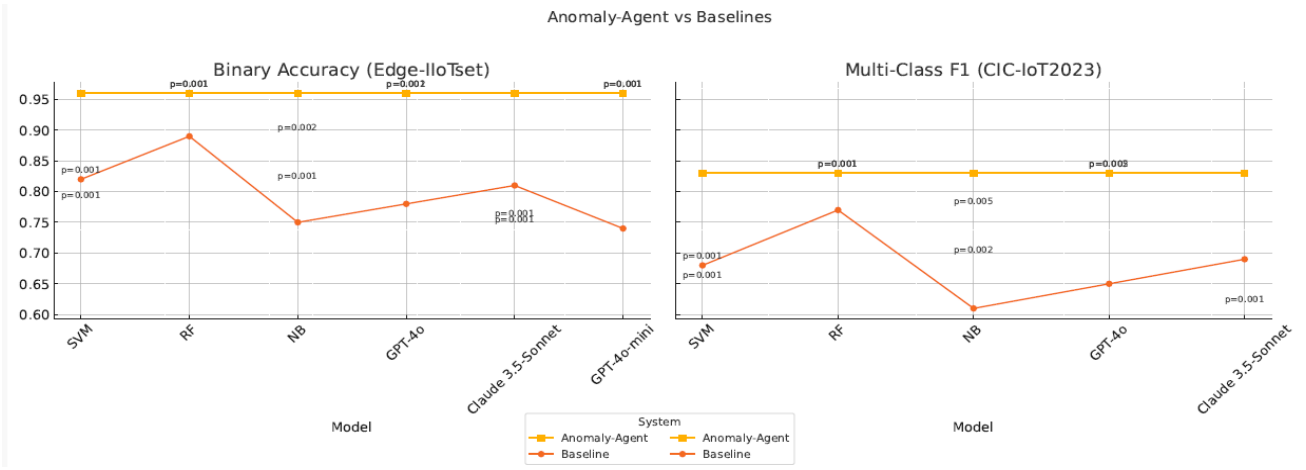


Fig. 8. Anomaly-Agent against baseline binary accuracy on Edge-IIoTset and multiclass $F1$ on CIC-IoT2023.

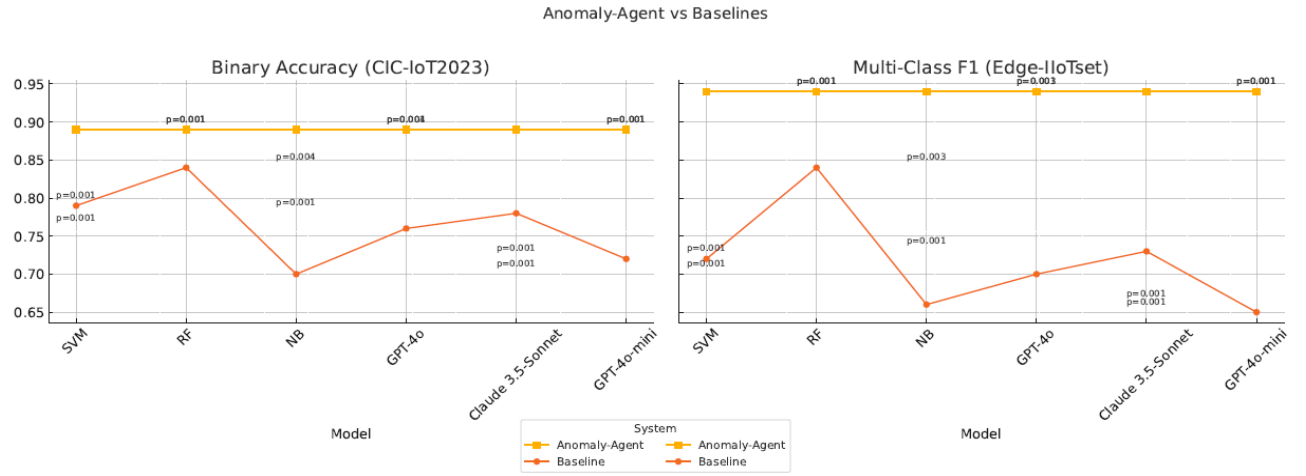


Fig. 9. Anomaly-Agent against baseline binary accuracy on CIC-IoT23 and multiclass $F1$ on Edge-IIoTset.

TABLE XIV
ANOMALY-AGENT COMPARISON WITH STATE-OF-THE-ART (SOTA) METHODS

Ref./Year	Model	Accuracy	Recall	Precision	F1-score	SHAP-XAI	LLM-Agent Integration
Ref.[30]	LSTM-CNN	99.4%,	99.5%	98.9%	99.3%	✗	✗
2024 Ref.[31]	MH-DRNN	96.89%,	98.5%	97.4%	97.6%	✗	✗
2024 Ref.[33]	SA-BiLSTM	96.54%	83.79%	90.94%	88.90%	✓	✗
2024 Ref.[34]	CIDIoT	77.20%,	57.53%	71.44%	63.73%	✓	✗
2024 **Ours**	**Anomaly-Agent**	**96.00%**	**94.00%**	**92.00%**	**93.00%**	✓	✓

V. CONCLUSION

This research presents Anomaly-Agent, the first LLM-powered agent for explainable anomaly detection in resource-constrained IoT/IIoT edge environments. By integrating dynamic reasoning, domain-specific tools, and SHAP-based interpretability, the agent overcomes key limitations of traditional ML and standalone LLM methods, particularly in handling zero-day attacks and providing

transparent decisions. The experimental evaluations on Edge-IIoTset and CIC-IoT2023 show Anomaly-Agent achieving 96% accuracy and 94% $F1$-score, significantly outperforming ML baselines, random forest: +12% $F1$, and LLMs GPT-4o: +26% recall. These improvements are statistically significant ($p < 0.01$), confirming robustness and reproducibility. The agent's explainability and tunable sensitivity make it a scalable, trustworthy solution for securing heterogeneous IoT/IIoT infrastructures. While our experiments validate scalability within controlled testbeds, real-world large-scale deployments may face communication bottlenecks, synchronization delays, or performance tradeoffs when operating across thousands of diverse devices. These challenges could limit the practicality of deploying the full Anomaly-Agent pipeline without hardware acceleration. In future work, we will explore federated learning to enable collaborative threat detection across distributed edge nodes while preserving privacy. Additionally, hardware acceleration using field programmable gate arrays (FPGAs) will be investigated to reduce SHAP-related latency for real-time deployment.

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